Brushless Driver SH-D08R

Content

1. Safety precautions.

In order to make you use this product safely, and to avoid harm and damage to you. Please fully understand the content before using this product. At the same time, without the consent of the company, do not use this product in life, safety, medical and other fields.

The following operations may damage the motor controller and motor.

- The rotating output shaft will not be touched during operation.
- Touch the motor output shaft with hands (Keyway, Incisor site).
- Handle the motor output shaft and cable during handling.
- The load is not firmly mounted on the motor output shaft.
- When assembling the motor and reducer, load fingers between the motor and reducer.
- If the emergency stop button or emergency protection circuit is not installed outside the equipment, the whole equipment can not be guaranteed to be in a safe state in case of equipment failure and abnormal action.

The following operations may cause burns.

- Touch the motor or driver in a short time during operation or after stopping.

The following operations may cause electric shock.

- The acceleration and deceleration time setter of the adjustment controller does not use insulating screws.
- The power supply for input and output signals does not use DC power supply with primary and secondary insulation.

The following operations may cause injury or damage to the equipment.

- Obstacles to ventilation are stacked around the motor controller.
- The motor controller was not firmly fixed on the mounting plate, resulting in falling.

The following operations may cause fire, electric shock or injury.

- Insert other objects into the opening of the controller.
- The operation is not stopped immediately when an exception occurs.
- When the motor exceeds the rated value of the controller.
- In normal operation, the surface temperature of the motor may exceed 70 °C.

2. Product series naming

◆ **Drive naming rule**

SH – D 08 H R

SH	D	08	H	R
Series	Type	Current	Voltage	Communication
SH: SH series	Driver	08: 8A	H: AC220V None: DC24-48V	R: RS-485

◆ **Naming rules of brushless dc motor**

M 30 L 3 2 A

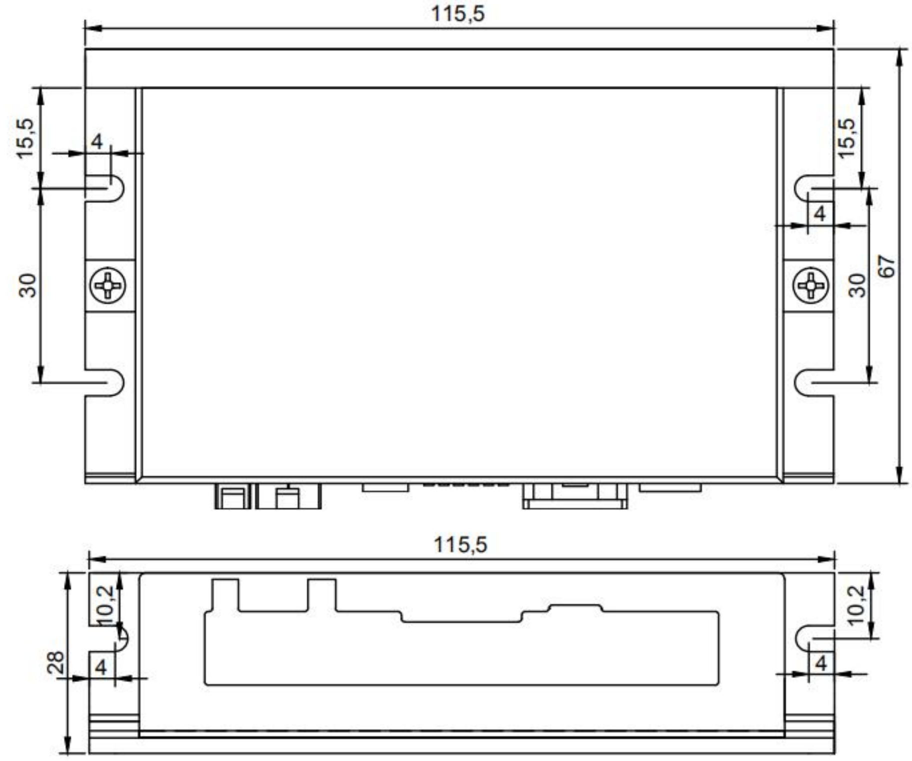
M	30	L	3	2	A
Type	Power	Voltage	Speed	Flange	Axle serial No.
Motor	30：30W、 60：60W、 90：90W、 120：120W	M：AC110 H：AC220 L：DC24 N：DC48	1：1000RPM 2：2000RPM 3：3000RPM	2：60MM 4：80MM 5：90MM	A：8 B：10 C：12 None：gearbox

◆ **Naming rules of gearbox**

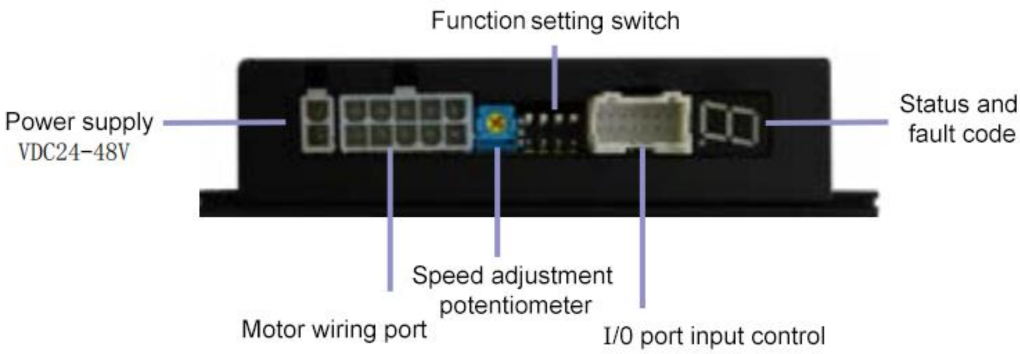
2 – G05

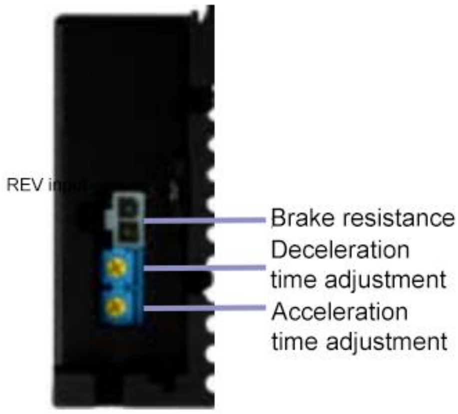
2	G05
Flange	Ratio
2：60MM 4：80MM 5：90MM	G05：1:5 G10：1:10 G20：1:20

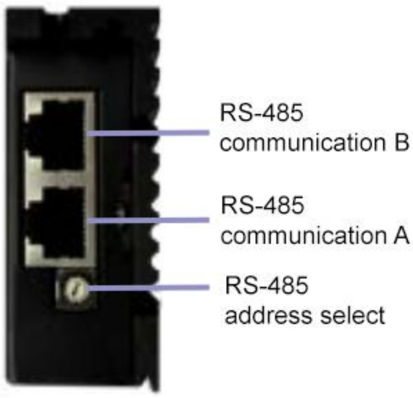
3. Driver size



4. Description of the drive port







5. Port and wiring

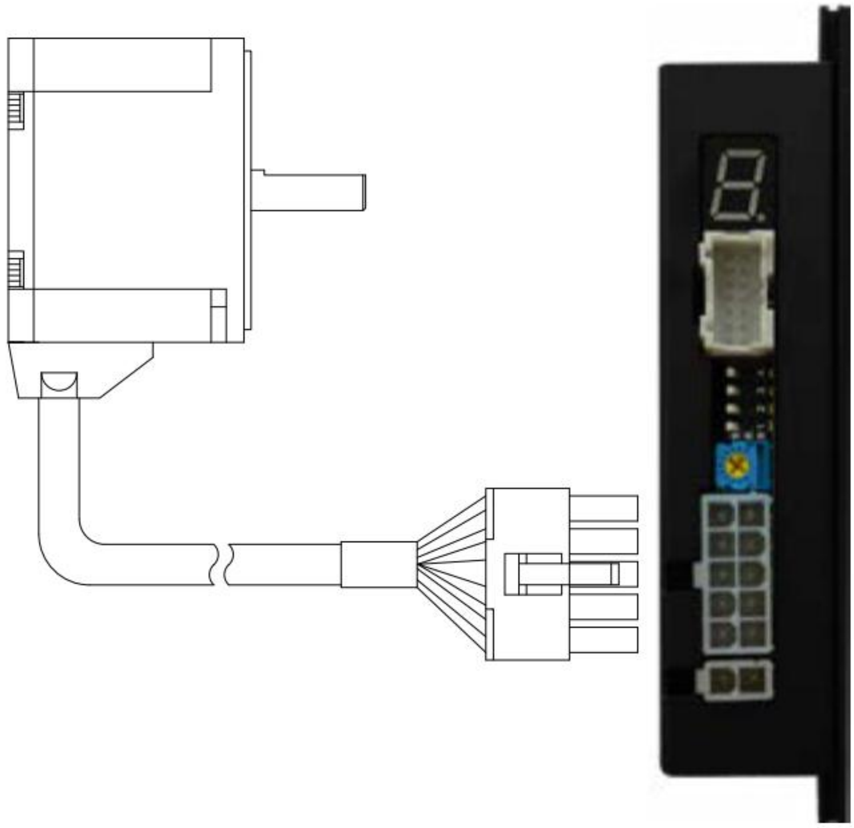
Connection of power supply

The DC+ is connected to the positive end of the power input, The DC- is connected to the negative end of the power supply.

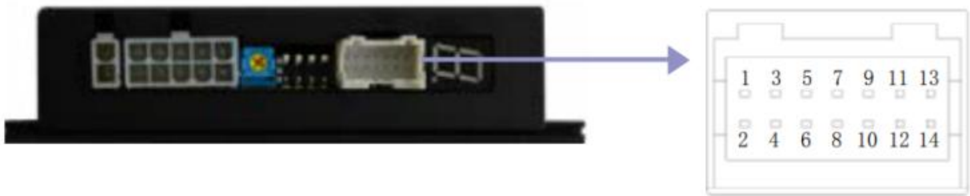
Connect the power cable to CN1

Input Voltage	Connect
DC24V ～ DC48V	DC+ is connected to the positive terminal, and DC- is negative terminal

Connection of motor.



Connection of input and output signals.



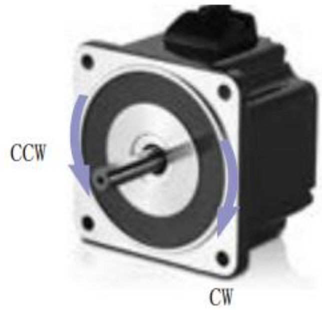
◆ **CN5 port:**

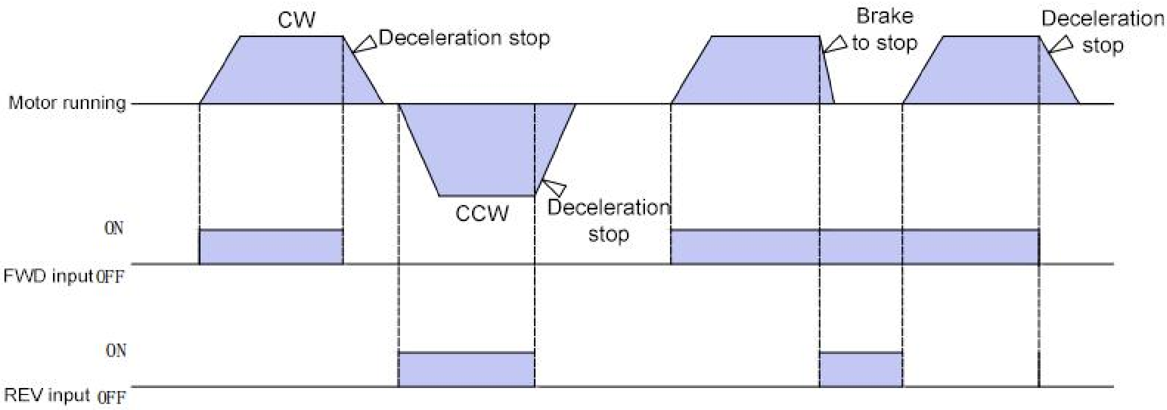
Function	Terminal lead color	Function description
5V	Black	5 V output, can be connected to potentiometer for external speed regulation.
SV	Black & White	External potentiometer interface, for external input analog speed regulation.
PWM	Blue	PWM(Duty cycle) input speed control, frequency speed control input interface.(Speed regulation mode C4 setting.)
GND	Blue & White	Control interface common terminal
FWD	Red	Controlling line for CW and connecting with GND is effective.
REV	Brown	Controlling line for CCW and connecting with GND is effective.
M0	Brown & White	Controlling line for multi speed and connecting with GND is effective.
M1	Brown & Black	~
M2	Orange	~
SPD	Red & White	Open drain output frequency, the motor pole is different, the output frequency will be different.
ALM	Green & White	Alarm output. In normal operation, the port is suspended, After the alarm, this port will be short circuited to the common terminal GND.
ALM-RST	Green	Alarm reset. Connecting this port to the GND will clear the fault alarm.
STB1	Yellow & Black	Maximum speed limit selection. En suspended, the maximum speed can be 4000 rpm. The port is connected to GND, and the maximum speed can be 10000 rpm.
STB2	Yellow	~

◆ Detailed input signal circuit

- **FWD/REV**

When FWD and common terminal are short circuited, the motor rotates clockwise; When FWD and common end (GND) are disconnected, the motor stops naturally;
When REV and common terminal (GND) are short circuited, the motor rotates anti-clockwise; When REV and common end are disconnected, the motor stops naturally;
FWD and REV are both disconnected from the common end (GND), the motor decelerates and stops;
FWD and REV are both connected with the common end (GND), the motor brake stops;





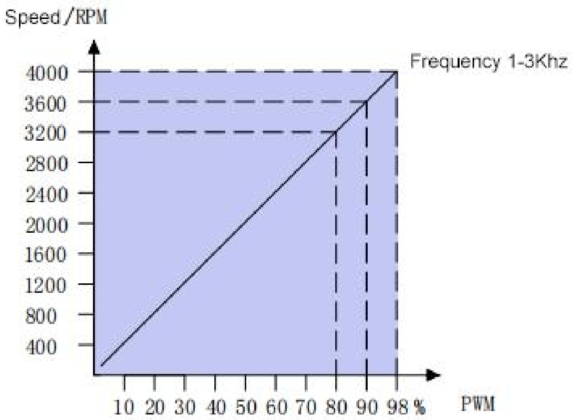
6. Function description

6.1 Speed control function selection(CN4)

"2"	"1"	Speed control mode
0	0	Internal potentiometer speed regulation
0	1	External analog and external potentiometer speed regulation.
1	0	PWM (Duty cycle) speed regulation.
1	1	Pulse frequency speed regulation.

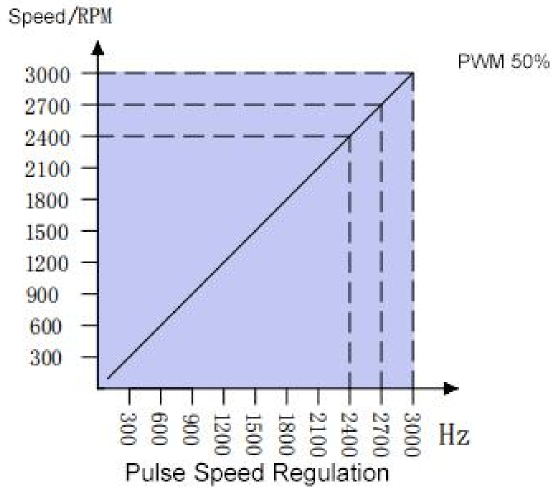
6.2 Introduction of speed regulation mode

A: PWM Speed control. When the CN4 is set as PWM, the speed will change with the duty ratio(1%-98%). The speed corresponding to the duty cycle will change according to the range of the actual speed setting.

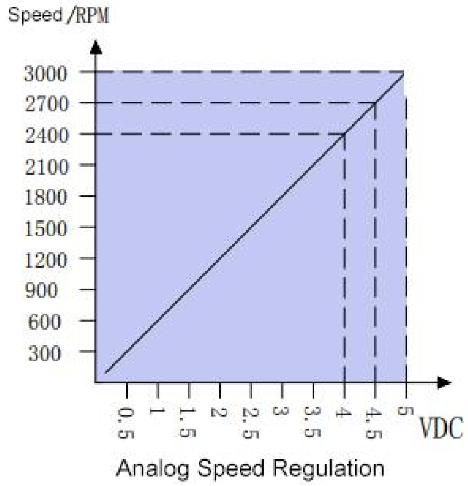


PWM Speed Regulation

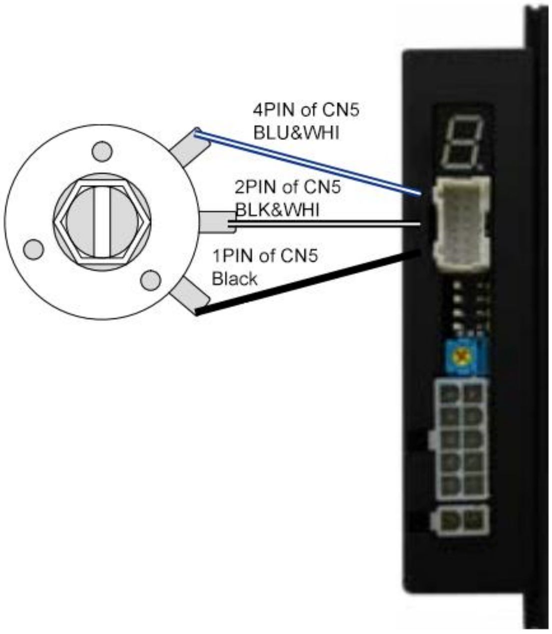
B: Pulse frequency speed regulation. When the CN4 is set as pulse frequency, the speed will change with the pulse frequency. (100HZ-3000HZ)

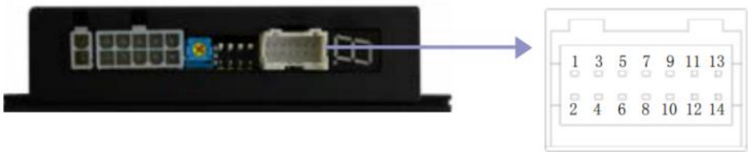


C: Analog input speed regulation. When the CN4 is set as analog input, the speed will change with the analog voltage(0-5VDC).



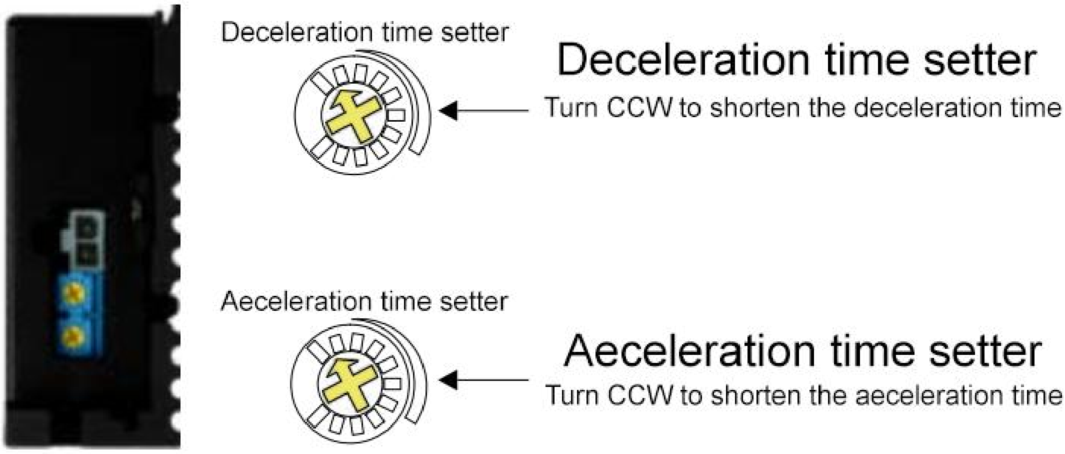
D: Speed regulation by external potentiometer. To use external potentiometer, set CN4 2=OFF 1=ON. The external potentiometer wiring is as follows:





6.3 Acceleration and deceleration time setting.

There are two adjusting knob on the side of the driver, they are used to adjust the acceleration and deceleration time.
The acceleration time is the time required from 0 to rated speed.
Deceleration time is the time required from rated speed to 0.



6.4 Brake resistor connection.

When the motor load inertia is large, connect the brake resistance to ensure the normal operation of the driver and the accuracy of the motor stop position.

The resistance should be selected according to the inertia. The larger the inertia, the smaller the resistance should be selected. The larger the inertia, the smaller the resistance should be chosen.

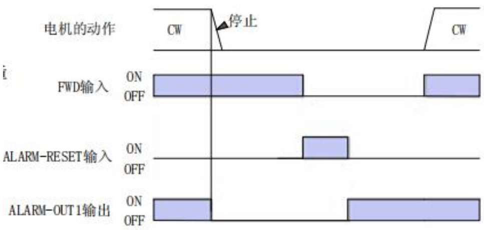
6.5 SPD Speed signal output.

The motor driver will detect pulse when the motor runs. The width of the pulse is 0.2ms. The speed of the electric appliance can be calculated according to formula below:

$$RPM = \frac{SPEED - OUT}{6 * Pole\ of\ pair} * 60$$

6.6 Alarm reset.

A: Connect ALM-RST and CN5 4(GND), the alarm will be relieved.



B: Turn off the power and restart after a minute.

6.7 M0 M1 M2 multi speed selection.

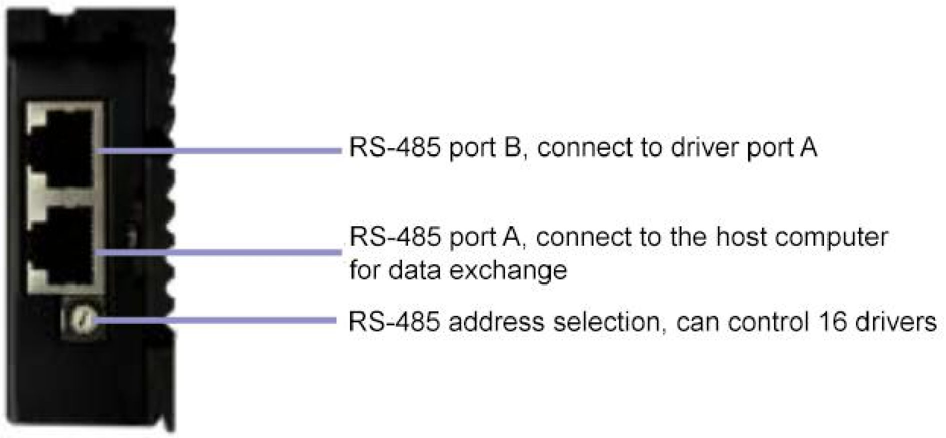
Seven speed can be selected via M0 M1 M2

M0	M1	M2	No	Speed
OFF	OFF	OFF	Other speed adjusting method	
ON	OFF	OFF	1	500
OFF	ON	OFF	2	1000
ON	ON	OFF	3	1500
OFF	OFF	ON	4	2000
ON	OFF	ON	5	2500
OFF	ON	ON	6	3000
ON	ON	ON	7	3500

6.8 RS-485 communication port

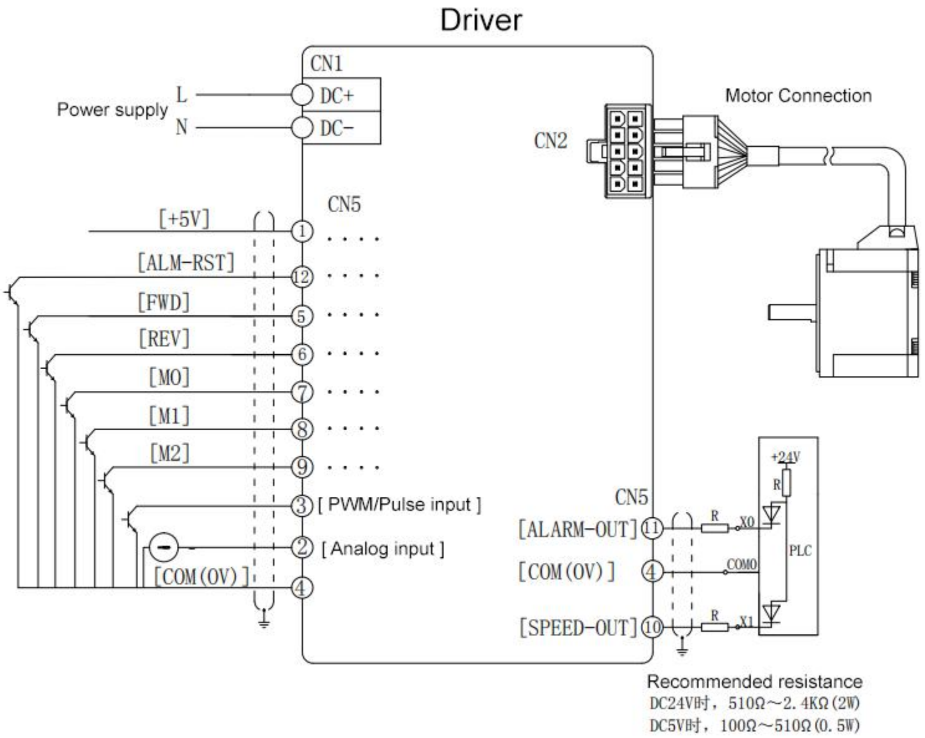
7.8 RS-485

The driver can be controlled by RS-485.

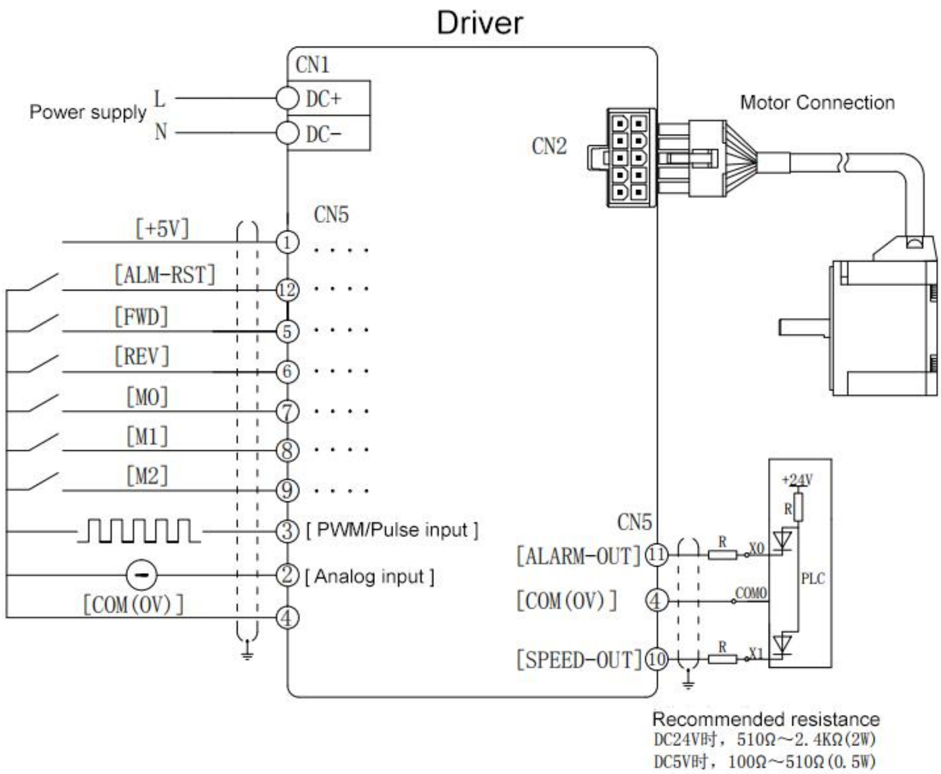


7. Motor driver connection diagram

A. Use PLC (transistor upper computer) to control motor

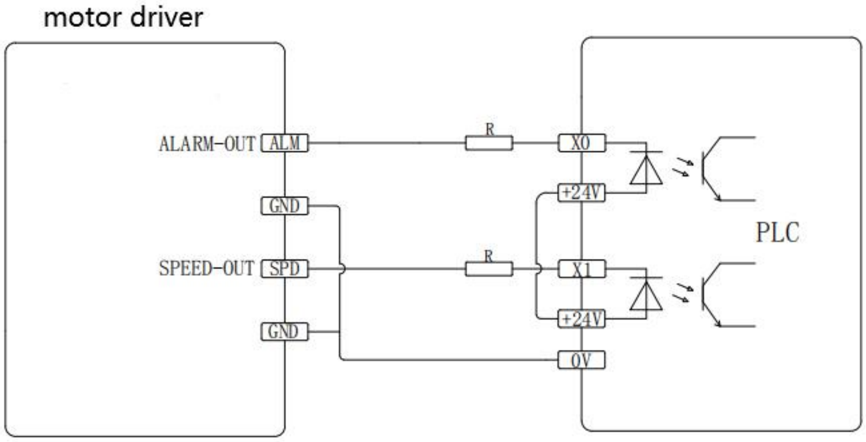


B. Use switch, relay to control motor

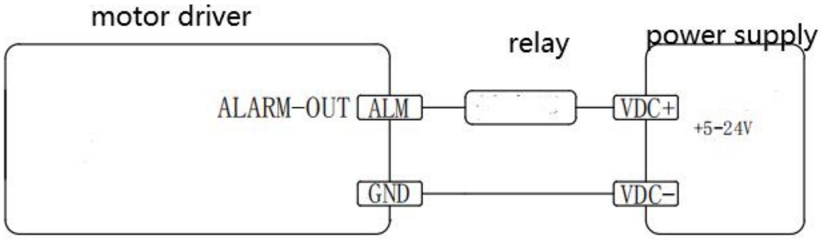


C. **Schematic diagram of output port wiring**.

PLC detecting output signal wiring.



D. Relay output wiring



8. Fault alarm description

Fault code	Fault name	Reason	Troubleshooting	Fault reset
1	Over current	A short circuit causes a large current input to the driver.	Confirm whether the circuit between the controller and the motor is damaged	Invalid
2	Over temperature	The driver's temperature is too high	Reduce the ambient temperature, Improve ventilation conditions.	Valid
3	Over voltage	Supply voltage reaches 130% of rated voltage	Reduce the load or the acceleration and deceleration time.	Valid
4	Low voltage	Supply voltage is 60% of rated voltage or lower	Make sure the power supply voltage is normal.	Valid
5	Sensor abnormal	The motor or sensor wire breaks or falls off.	Please confirm the sensor wire.	Valid
6	Over speed	The motor speed exceeds 4800 rpm		Valid
7	Locked rotor protection	The external load is too large in a moment.	Check the load	Valid
8	System error	There is a fault in the control system circuit.	Please contact technical support.	Invalid

9. Fault diagnosis and troubleshooting

When the motor can not operate normally, please refer to the following troubleshooting. If it can't be solved still, please contact technical support inquiry@ican-tech.com

Situation	Trouble Shooting	Approach
Motor do not rotate	Incorrect power connection	Please confirm power supply wiring
	[External signal input] When the parameter is invalid, FWD input or REV input is set to ON	Please set input signal to OFF, set [External signal input] to valid
	FWD and REV input are OFF	Set one of them to ON
	FWD and REV input are ON	
	Alarm	The protection function works and an Alarm occurs. Resolve after investigating the cause
Opposite direction of rotation	FWD and REV are connected reversely, or the connection is incorrect	Please confirm FWD and REV wiring
	Use conjoined parallel shaft reducer, ratio is 30/50/100	The rotation direction of the output shaft of these reducers is opposite to that of the motor output shaft. Please reverse FWD and REV
	The switch of the direction of rotation is set wrong	Please confirm the setting of rotation switch
Speed cannot be increased	Set speed limit	Please set the speed limit to 4000r/min
The motor movement is unstable and the vibration is too large	The output shaft of the motor (gearbox) is not aligned with the load shaft	Please confirm the connection status
	Affected by interference	Please confirm the operation status when using motor,driver and external equipment. If it is affected by interference, take the following measures: • Isolate the source of interference. • Adjust the wiring. • Change the signal cable to shielded cable. • Install the ferrite core.

10. After sale service

10.1 Warranty period

Dongguan ICAN Technology provides warranty for 1 year from the date of shipping.

10.2 Return policy

1. After-use or man-made damage condition (etc, wrong wiring), no return
2. ICAN Technology guarantees the product quality, but product incompatibility is not in the return or maintain condition.
3. Customers don't use the products under the specified electrical performance and environment indicators, no maintain condition
4. Customers change the internal parts

10.3 Maintenance process

1) Get the maintenance permission

2) Ship the package to the following address: 4/F, Block B, RuiLian Zhenxing Industrial Park, Wanjiang District, Dongguan City, Guangdong Province

Tel: 86-0769-22327568

Web: www.ican-motor.com

Email: Inquiry@ican-tech.com